



Caringbah High School

Year 12 2022

Mathematics Extension 2

HSC Course

Assessment Task 4 – Trial HSC Examination

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black or blue pen
- NESA-approved calculators may be used
- A reference sheet is provided
- In Questions 11-16, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 100

Section I 10 marks

Attempt Questions 1-10

Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 90 marks

Attempt Questions 11-16

Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.

Name: _____

Class: _____

Marker's Use Only							
Section I	Section II						Total
Q 1-10	Q11	Q12	Q13	Q14	Q15	Q16	
/10	/15	/15	/15	/15	/15	/15	/100

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet (page 13) for Questions 1–10

1. Let the point P on an Argand diagram represent the complex number z . After being multiplied by another complex number, ω , P is rotated 90° anti-clockwise and z is enlarged by a factor of 3. Which of the following is the value of ω ?
- (A) $3i$
(B) $-3i$
(C) e^{-3i}
(D) $3e^{\frac{\pi}{2}i}$
2. A particle moves in a straight line so that its displacement, x metres, at any time, t seconds, is given by $x = 5 \sin 3t + 12 \cos 3t$. What is the speed of the particle as it passes through the centre of its motion?
- (A) 12 m/s
(B) 13 m/s
(C) 39 m/s
(D) 117 m/s
3. Which of the following is equivalent to the expression $\frac{12x-3}{(x-2)(x^2-3x+2)}$?
- (A) $\frac{9}{x-1} + \frac{9}{x-2} - \frac{21}{(x-2)^2}$
(B) $\frac{9}{x-1} + \frac{18}{x-2} - \frac{21}{(x-2)^2}$
(C) $\frac{9}{x-1} - \frac{9}{x-2} + \frac{21}{(x-2)^2}$
(D) $\frac{9}{x-1} - \frac{18}{x-2} + \frac{21}{(x-2)^2}$

4. Which of the following is the vector equation of a line that passes through the

point $(1, 3, -2)$ and is perpendicular to the line $\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$?

(A) $\begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$

(B) $\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$

(C) $\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix}$

(D) $\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ -3 \end{pmatrix}$

5. The acceleration of a particle is given by $a = e^{-3t}$ m/s² where the particle has an initial velocity of 2 m/s. What is the terminal velocity of the particle?

(A) $1.\dot{3}$ m/s

(B) $1.\dot{6}$ m/s

(C) $2.\dot{3}$ m/s

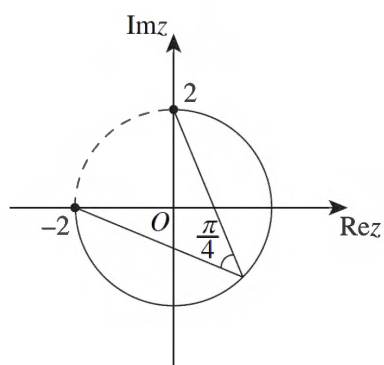
(D) $2.\dot{6}$ m/s

Section I continues on page 5

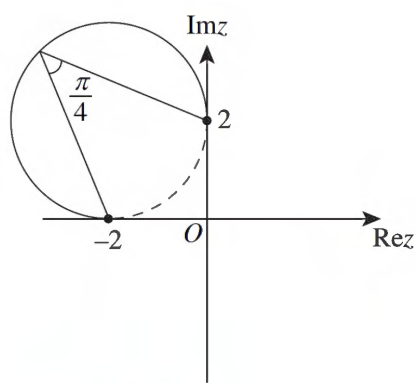
6. Which of the following diagrams best represents the solutions to the equation

$$\arg\left(\frac{z-2}{z+2i}\right) = \frac{\pi}{4}?$$

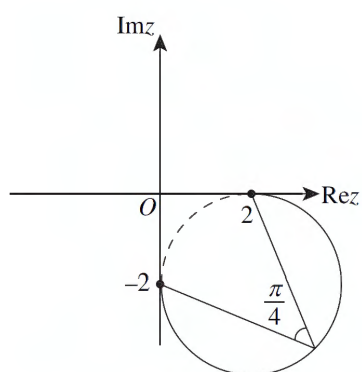
(A)



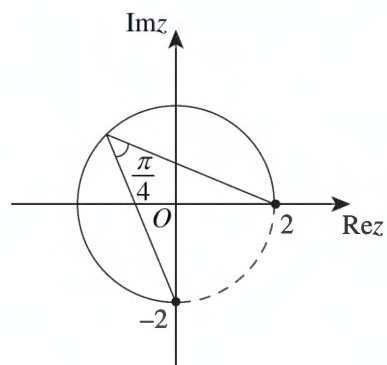
(B)



(C)



(D)



7. The points A , B and C are collinear where $\overrightarrow{OA} = \underline{i} - \underline{j}$, $\overrightarrow{OB} = -3\underline{j} - \underline{k}$ and $\overrightarrow{OC} = 2\underline{i} + a\underline{j} + b\underline{k}$ for some constants a and b . What are the values of a and b ?
- (A) $a = -1$ and $b = -1$
 (B) $a = -1$ and $b = 1$
 (C) $a = 1$ and $b = -1$
 (D) $a = 1$ and $b = 1$
8. Consider the statement: $\exists x \in \mathbb{R}, \ln x = 1 \text{ and } x > 2$
 Which of the following is the negation of the statement?
- (A) $\exists x \in \mathbb{R}, \ln x \neq 1 \text{ or } x \leq 2$
 (B) $\exists x \in \mathbb{R}, \ln x \neq 1 \text{ and } x \leq 2$
 (C) $\forall x \in \mathbb{R}, \ln x \neq 1 \text{ or } x \leq 2$
 (D) $\forall x \in \mathbb{R}, \ln x \neq 1 \text{ and } x \leq 2$
9. If $\int_1^4 f(x) dx = k$ for some constant k , what is the value of $\int_1^4 f(5-x) dx$?
- (A) $-k$
 (B) $5-k$
 (C) $k+5$
 (D) k
10. Which of the following best describes the path of a particle with the following parametric equations?
- $$\begin{cases} x = t \sin t \\ y = t \cos t \\ z = t \end{cases}$$
- (A) spiral around the z -axis, traversing in a clockwise direction
 (B) spiral around the z -axis, traversing in an anticlockwise direction
 (C) helix around the z -axis, traversing in a clockwise direction
 (D) helix around the z -axis, traversing in an anticlockwise direction

End of Section I

Section II

60 marks

Attempt Questions 11–16

Allow about 2 hours 45 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Consider the complex numbers $z = 2 + 2\sqrt{3}i$ and $w = 2\sqrt{3} - 2i$.
- (i) Find z and w in modulus-argument form. 2
 - (ii) Find zw and $\frac{z}{w}$ in modulus-argument form. 2
 - (iii) Describe the relationship between z and w geometrically. 1
- (b) Consider the equation $f(z) = 0$ where $f(z) = z^3 - 11z^2 + 55z - 125$.
- (i) Find the three roots of the equation in the form $a + ib$, where a and b are real. 3
 - (ii) Show that the points A , B and C in the Argand diagram representing these roots lie on a circle of the form $|z| = k$ for some constant k , and find the area of $\triangle ABC$. 2
- (c) (i) Find constants A and B such that 2
- $$A(3 \sin x + 2 \cos x) + B(3 \cos x - 2 \sin x) = 8 \sin x + 14 \cos x$$
- (ii) Hence, find the exact value of $\int_0^{\frac{\pi}{2}} \frac{8 \sin x + 14 \cos x}{3 \sin x + 2 \cos x} dx$ 3

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) We are given that $\frac{1}{a}$, $\frac{1}{b}$ and $\frac{1}{c}$ are three consecutive terms in a geometric series, where $a, b, c \in \mathbb{R}$. Show that $a^2 + c^2 \geq ab + bc$. 3
- (b) A body of unit mass falls under gravity through a resisting medium. The body falls from rest from a height above the ground. The resistance to its motion is $\frac{1}{100}v^2$ where v m/s is the speed of the body when it has fallen a distance x m. The acceleration due to gravity is g m/s².
- (i) Show that the equation of motion of the body is $\ddot{x} = g - \frac{1}{100}v^2$ 1
- (ii) Show that the terminal velocity V m/s of the body is given by $V = 10\sqrt{g}$ 1
- (iii) Hence show that $v^2 = V^2 \left(1 - e^{-\frac{x}{50}}\right)$ 3
- (iv) Find the distance fallen in metres until the body reaches a velocity equal to half of its terminal velocity. 2
(You may assume this occurs before the body reaches the ground.)
- (c) (i) Use the substitution $u = \frac{1}{x}$ to show that $\int_{\frac{1}{a}}^a \frac{\ln x}{1+x^2} dx = 0$ 2
for any constant $a > 0$.
- (ii) Hence, find $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\tan^{-1} x}{x} dx$ in simplest exact form. 3

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) Consider the lines L_1 and L_2 , determined by the vector equations

$$L_1: \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad \text{and} \quad L_2: \mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

Show that L_1 and L_2 intersect and are perpendicular, stating the coordinates of the point of intersection. 3

- (b) A body of mass m kg is travelling in a horizontal straight line so that the resultant force on the body is a resistance force of magnitude $\frac{1}{10}m\sqrt{1+v}$ when its speed is v m/s. Initially, the speed of the body is 15 m/s.

- (i) Find the time taken for the body to come to rest. 2
(ii) Find the distance travelled by the body in coming to rest. 3

- (c) (i) Find the real numbers A and B such that

$$\frac{5}{(x+3)(2x+1)} = \frac{A}{x+3} + \frac{B}{2x+1} \quad 2$$

- (ii) Hence, or otherwise, evaluate

$$\int_0^2 \frac{5}{(x+3)(2x+1)} dx \quad 2$$

- (d) Find the points of intersection between the sphere S and the line L , given below. 3

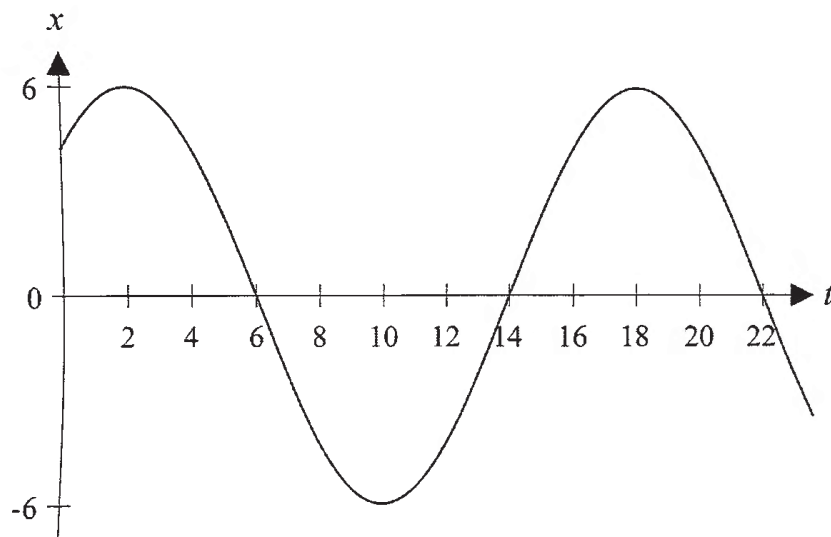
$$S: \left| \mathbf{r} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right| = 3$$

$$L: \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) The graph below shows the displacement x cm from the centre of motion at time t seconds for a particle performing simple harmonic motion in a straight line.

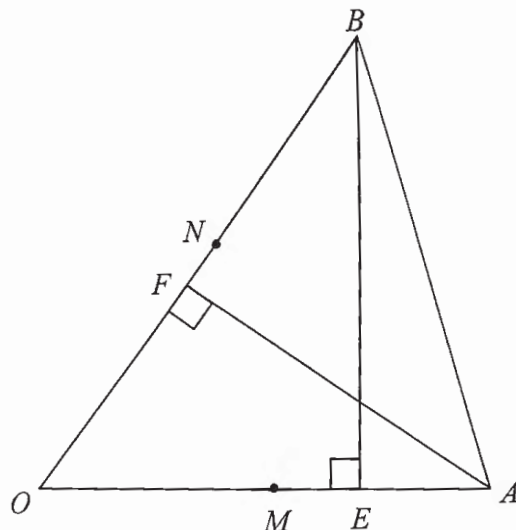


- (i) Express displacement as a function of time in the form $x = A \sin(nt + \alpha)$ 2
- (ii) Find the distance travelled by the particle in the first 30 seconds of its motion after observation began at time $t = 0$. 2
- (b) (i) If $t = \tan \theta$, prove that $\tan 4\theta = \frac{4t(1-t^2)}{1-6t^2+t^4}$ 2
- (ii) If $\tan \theta \tan 4\theta = 1$, deduce that $5t^4 - 10t^2 + 1 = 0$ 2
- (iii) Given that $\theta = \frac{\pi}{10}$ and $\theta = \frac{3\pi}{10}$ are roots of the equation $\tan \theta \tan 4\theta = 1$, find the exact value of $\tan \frac{\pi}{10}$. 2
- (c) (i) Show that for $k \geq 2$, $\tan^{-1}\left(\frac{1}{k-1}\right) - \tan^{-1}\left(\frac{1}{k+1}\right) = \tan^{-1}\left(\frac{2}{k^2}\right)$ 2
- (ii) Hence, show that $\lim_{n \rightarrow \infty} \left[\sum_{k=1}^n \tan^{-1}\left(\frac{2}{k^2}\right) \right] = \frac{3\pi}{4}$ 3

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) In $\triangle OAB$, BE is the altitude from B to OA and AF is the altitude from A to OB . M and N are the midpoints of OA and OB respectively. $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$.



Not to scale

Use vector methods to show that $|OM||OE| = |OM||OF|$

3

- (b) Let $I_n = \int_0^1 \frac{x^n}{x^2+1} dx$ for $n \geq 2$

- (i) Show that $I_n + I_{n-2} = \frac{1}{n-1}$

2

- (ii) Hence, or otherwise, show that $\int_0^1 \frac{x^2(x-1)^2}{x^2+1} dx = \ln 2 - \frac{2}{3}$

3

- (c) Consider the functions $f_n(x) = e^x - \left(1 + \sum_{r=1}^n \frac{x^r}{r!}\right)$, $n = 1, 2, 3, \dots$

- (i) Show that $f_n(0) = 0$ and $f_{n+1}'(x) = f_n(x)$ for $n = 1, 2, 3, \dots$

2

- (ii) Show that $f_1(x) > 0$ for all $x > 0$ and hence $1+x < e^x$ for all $x > 0$

2

- (iii) Use mathematical induction to show that for all positive integers $n \geq 1$,

$$1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} < e^x \text{ for all } x > 0$$

3

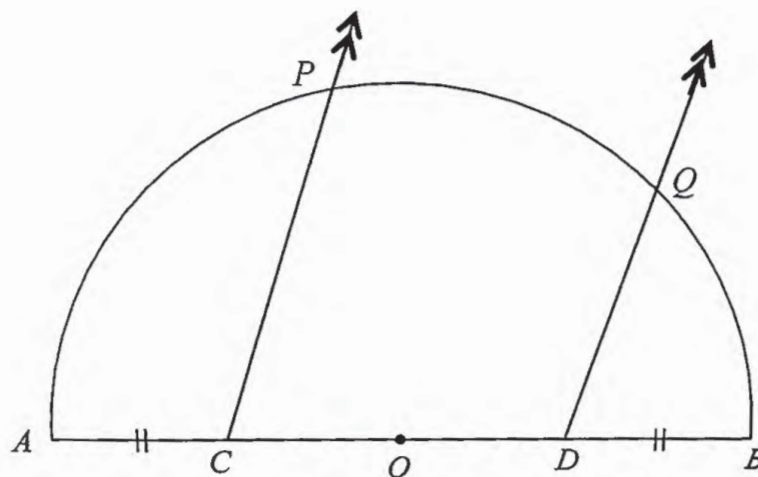
End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

(a) (i) Show that $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta}$, where $\cos \theta \neq 0$, $n \in \mathbb{Z}^+$ 2

(ii) Hence, show that the roots of $(1 + z)^2 + (1 - z)^2 = 0$ are $z = \pm i \tan \frac{\pi}{4}$ when z is purely imaginary. 2

- (b) A semi-circle is drawn on diameter AB . O is the midpoint of AB and the points C and D lie on AB such that $AC = BD$. Parallel lines are drawn through C and D , intersecting the semi-circle at P and Q respectively. $\overrightarrow{OC} = \underline{c}$ and $\overrightarrow{CP} = \underline{p}$.



(i) Explain why $\overrightarrow{DQ} = \lambda \underline{p}$ for some scalar $\lambda > 0$, then show that $(1 - \lambda) \underline{p} \cdot \underline{p} + 2 \underline{c} \cdot \underline{p} = 0$ 3

(ii) Hence show that $\angle CPQ = 90^\circ$. 2

(c) (i) For integer values of n , $I_n = \int_1^e (1 - \ln x)^n dx$ for $n \geq 0$. Show that $I_n = -1 + nI_{n-1}$ for $n \geq 1$. 2

(ii) Use mathematical induction to show that $I_n = n!e - 1 - \sum_{r=1}^n {}^nP_r$ for all integers $n \geq 1$ 4

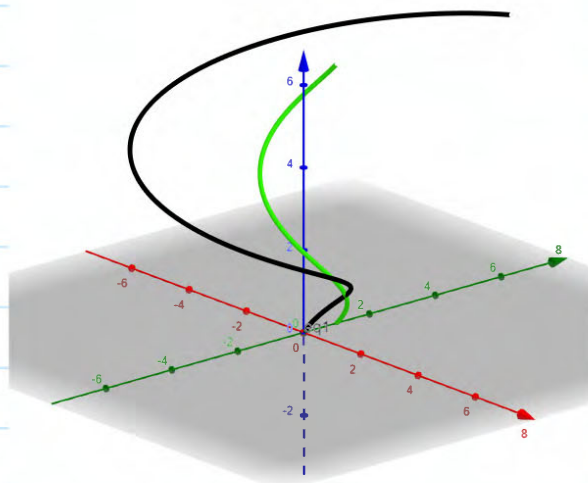
End of Examination

Section 1

- | | | | | |
|------|------|------|------|-------|
| 1. A | 2. C | 3. C | 4. C | 5. C |
| 6. D | 7. D | 8. C | 9. D | 10. A |

Section 2

Question 11



(a) $z = 2 + 2\sqrt{3}i$ $w = 2\sqrt{3} - 2i$

(i) $|z| = \sqrt{2^2 + (2\sqrt{3})^2} = 4$

$\arg(z) = \tan^{-1}\left(\frac{2\sqrt{3}}{2}\right) = \frac{\pi}{3}$

$\therefore z = 4\text{cis}\frac{\pi}{3}$ ✓

$|w| = \sqrt{(2\sqrt{3})^2 + (-2)^2} = 4$

$\arg(w) = \tan^{-1}\left(\frac{-2}{2\sqrt{3}}\right) = -\frac{\pi}{6}$

$\therefore w = 4\text{cis}\left(-\frac{\pi}{6}\right)$ ✓

(ii) $zw = 4\text{cis}\frac{\pi}{3} \times 4\text{cis}\left(-\frac{\pi}{6}\right)$

$= 16\text{cis}\frac{\pi}{6}$ ✓

$\frac{z}{w} = \frac{4\text{cis}\frac{\pi}{3}}{4\text{cis}\left(-\frac{\pi}{6}\right)}$

$= \text{cis}\frac{\pi}{2}$ ✓

(iii) $\frac{z}{w} = \text{cis}\frac{\pi}{2} = i \Rightarrow z = iw$ or $w = -iz$

$\therefore z$ is w rotated anti-clockwise 90° ✓
or w is z rotated clockwise 90°

$$(b) f(z) = z^3 - 11z^2 + 55z - 125 = 0$$

$$(i) \text{ Test } f(5) = 5^3 - 11(5)^2 + 55(5) - 125 = 0$$

$\therefore z=5$ is a root $\Rightarrow (z-5)$ is a factor ✓

$$\begin{array}{r} z^2 - 6z + 25 \\ z-5 \overline{) z^3 - 11z^2 + 55z - 125} \\ \underline{z^3 - 5z^2} \\ -6z^2 + 55z \\ \underline{-6z^2 + 30z} \\ 25z - 125 \\ \underline{25z - 125} \\ 0 \end{array}$$

$$\therefore f(z) = (z-5)(z^2 - 6z + 25) \quad \checkmark$$

$$z = \frac{6 \pm \sqrt{-64}}{2(1)} = \frac{6 \pm 8i}{2} = 3 \pm 4i$$

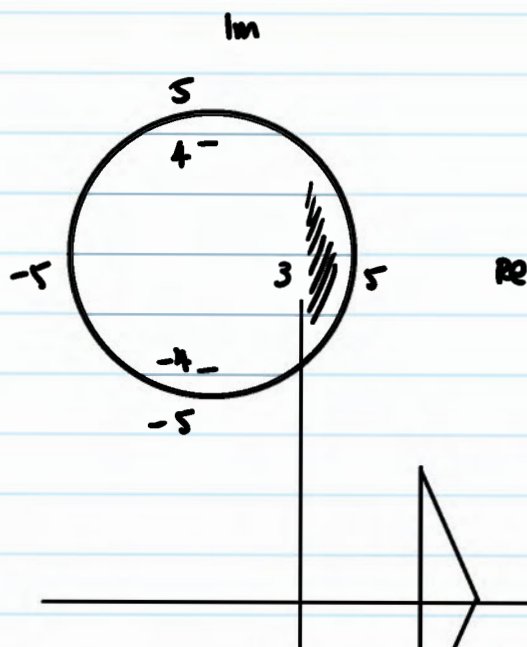
$$\therefore z = 5, 3 \pm 4i \quad \checkmark$$

$$(ii) |5| = 5$$

$$|3+4i| = \sqrt{3^2 + 4^2} = 5$$

$$|3-4i| = \sqrt{3^2 + (-4)^2} = 5$$

$\therefore$ Roots lie on circle $|z|=5$ ✓



$$\text{Area} = \frac{1}{2}bh$$

$$= \frac{1}{2} \times 2 \times 8$$

$$= 8 \text{ units}^2 \quad \checkmark$$

$$(c) (i) \quad A(3\sin x + 2\cos x) + B(3\cos x - 2\sin x) = 8\sin x + 14\cos x$$

$$3A\sin x + 2A\cos x + 3B\cos x - 2B\sin x = 8\sin x + 14\cos x$$

$$(3A - 2B)\sin x + (2A + 3B)\cos x = 8\sin x + 14\cos x$$

Equating coefficients,

$$3A - 2B = 8 \quad \text{--- (1)}$$

$$2A + 3B = 14 \quad \text{--- (2)}$$

$$(1) \times 2: \quad 6A - 4B = 16 \quad \text{--- (3)}$$

$$(2) \times 3: \quad 6A + 9B = 42 \quad \text{--- (4)}$$

$$(4) - (3): \quad 13B = 26$$

$$B = 2 \quad \text{--- (5)}$$

$$\text{Sub (5) in (1): } 3A - 2(2) = 8$$

$$3A = 12$$

$$A = 4$$

$$\therefore A = 4, B = 2$$

$$(ii) \quad \int_0^{\pi/2} \frac{8\sin x + 14\cos x}{3\sin x + 2\cos x} dx$$

$$= \int_0^{\pi/2} \frac{4(3\sin x + 2\cos x) + 2(3\cos x - 2\sin x)}{3\sin x + 2\cos x} dx$$

$$= \int_0^{\pi/2} 4 + \frac{2(3\sin x + 2\cos x)'}{3\sin x + 2\cos x} dx$$

$$= \left[4x + 2\ln|3\sin x + 2\cos x| \right]_0^{\pi/2}$$

$$= \frac{4\pi}{2} + 2\ln|3(1) + 2(0)| - (0 + 2\ln|3(0) + 2(1)|)$$

$$= 2\pi + 2\ln(3/2)$$

Question 12

(a) GP: $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ $a, b, c \in \mathbb{R}$

Common ratio $r = \frac{\frac{1}{b}}{\frac{1}{a}} = \frac{\frac{1}{c}}{\frac{1}{b}}$

$$\frac{a}{b} = \frac{b}{c}$$

$$\therefore ac = b^2 \text{ --- ①}$$

Now, consider $(a-b)^2 = a^2 - 2ab + b^2 \geq 0$

$$a^2 + b^2 \geq 2ab \text{ --- ②}$$

Similarly, $b^2 + c^2 \geq 2bc \text{ --- ③}$

$$a^2 + c^2 \geq 2ac \text{ --- ④}$$

$$\text{②} + \text{③} + \text{④}: 2(a^2 + b^2 + c^2) \geq 2(ab + bc + ac)$$

$$a^2 + b^2 + c^2 \geq ab + bc + ac \text{ from ①}$$

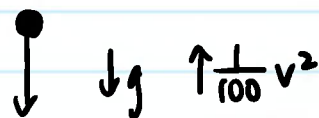
$$\therefore a^2 + c^2 \geq ab + bc, \text{ as required}$$

(b)(i) $F = ma$

$$g - \frac{1}{100}v^2 = 1 \times \ddot{x}$$

$$\therefore \ddot{x} = g - \frac{1}{100}v^2$$

✓ Show



(ii) When $\ddot{x} = 0$, $v = V > 0$

$$0 = g - \frac{1}{100}V^2$$

$$\frac{1}{100}V^2 = g$$

$$V^2 = 100g$$

$$\therefore V = 10\sqrt{g}$$

✓ Show

$$(iii) \quad \ddot{x} = v \frac{dv}{dx} = g - \frac{1}{100} v^2$$

$$\frac{dv}{dx} = \frac{100g - v^2}{100v}$$

$$\frac{dx}{dv} = \frac{100v}{100g - v^2}$$

$$x = \int \frac{100v}{100g - v^2} dv$$

$$= -50 \ln |100g - v^2| + C$$

$$\text{At } t=0, x=0, v=0$$

$$0 = -50 \ln |100g - 0^2| + C$$

$$C = 50 \ln 100g$$

$$\therefore x = 50 \ln \left| \frac{100g}{100g - v^2} \right|$$

$$\frac{x}{50} = \ln \left| \frac{100g}{100g - v^2} \right|$$

$$e^{\frac{x}{50}} = \frac{100g}{100g - v^2}$$

$$100g - v^2 = 100g e^{-x/50}$$

$$v^2 = 100g (1 - e^{-x/50})$$

$$\therefore v^2 = V^2 (1 - e^{-x/50})$$

$$(iv) \quad \text{Sub } v = \frac{V}{2}$$

$$\frac{V^2}{4} = V^2 (1 - e^{-x/50})$$

$$e^{-x/50} = \frac{3}{4}$$

$$-\frac{x}{50} = \ln \frac{3}{4}$$

$$x = 50 \ln \frac{4}{3}$$

$$\therefore x = 14.38 \text{ m (2dp)}$$

$$(c) \int_{\frac{1}{a}}^a \frac{\ln x}{1+x^2} dx$$

$$= \int_a^{\frac{1}{a}} \frac{\ln(\frac{1}{u})}{1+(\frac{1}{u})^2} \cdot \frac{-1}{u^2} du$$

$$= \int_{\frac{1}{a}}^a \frac{-\ln u}{\frac{u^2+1}{u^2}} \cdot \frac{1}{u^2} du \quad \checkmark$$

$$= \int_{\frac{1}{a}}^a \frac{-\ln u}{1+u^2} du$$

$$= \int_{\frac{1}{a}}^a \frac{-\ln x}{1+x^2} dx \quad \text{Dummy variable}$$

$$2 \int_{\frac{1}{a}}^a \frac{\ln x}{1+x^2} dx = 0 \quad \checkmark$$

$$\therefore \int_{\frac{1}{a}}^a \frac{\ln x}{1+x^2} dx = 0$$

Let $u = \frac{1}{x} \rightarrow x = \frac{1}{u}$
 $du = -\frac{1}{x^2} dx$
 $dx = -x^2 du$
 $dx = -\frac{1}{u^2} du$
 $\begin{cases} x = \frac{1}{a}, & u = a \\ x = a, & u = \frac{1}{a} \end{cases}$

$$(ii) \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\tan^{-1} x}{x} dx$$

$u = \tan^{-1} x \quad v' = \frac{1}{x}$
 $u' = \frac{1}{1+x^2} \quad v = \ln x$

$$= \left[\ln x \cdot \tan^{-1} x \right]_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} - \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\ln x}{1+x^2} dx \quad \checkmark$$

$$= \ln \sqrt{3} \cdot \frac{\pi}{3} - \ln\left(\frac{1}{\sqrt{3}}\right) \cdot \frac{\pi}{6} - \underline{0} \quad \text{from (i)} \quad \checkmark$$

$$= \frac{\pi \ln \sqrt{3}}{3} + \frac{\pi \ln \sqrt{3}}{6}$$

$$= \frac{\pi \ln \sqrt{3}}{2}$$

$$= \frac{\pi \ln 3}{4} \quad \checkmark$$

Question 13

(a) $L_1: \underline{r} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$ and $L_2: \underline{r} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$

$$\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = 2 \times 1 + -1 \times 1 + 1 \times -1 = 0$$

$$\therefore L_1 \perp L_2$$

Then, POI when $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$

$$3 + 2\lambda = -1 + \mu \quad \text{--- (1)}$$

$$2 - \lambda = 1 + \mu \quad \text{--- (2)}$$

$$-1 + \lambda = -\mu \quad \text{--- (3)}$$

$$\textcircled{1} - \textcircled{2}: 1 + 3\lambda = -2$$

$$\lambda = -1 \quad \text{--- (4)}$$

Sub (4) in (2): $2 - (-1) = 1 + \mu$

$$\mu = 2$$

Test $\lambda = -1, \mu = 2$ in (3): LHS = $-1 - 1 = -2$

$$\text{RHS} = -2$$

$$\therefore \text{POI at } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} - 1 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$$

$$(b)(i) \quad F = ma$$

$$\frac{-1}{10} \sqrt{1+v} = m \ddot{x}$$

$$\ddot{x} = \frac{dv}{dt} = \frac{-1}{10} \sqrt{1+v}$$

$$\frac{dt}{dv} = \frac{-10}{\sqrt{1+v}} \quad \checkmark$$

$$\int_0^T dt = \int_{15}^0 -10(1+v)^{-1/2} dv$$

$$[t]_0^T = 10 \left[\frac{(1+v)^{1/2}}{1/2} \right]_0^{15}$$

$$T - 0 = 20(\sqrt{16} - \sqrt{1})$$

$$\therefore T = 60 \text{ seconds} \quad \checkmark$$

$$(ii) \quad \ddot{x} = v \frac{dv}{dx} = \frac{-1}{10} \sqrt{1+v}$$

$$\frac{dv}{dx} = \frac{-\sqrt{1+v}}{10v}$$

$$\frac{dx}{dv} = \frac{-10v}{\sqrt{1+v}} \quad \checkmark$$

$$\int_0^D dx = \int_{15}^0 \frac{-10v}{\sqrt{1+v}} dv$$

$$[x]_0^D = 10 \int_0^{15} \frac{1+v-1}{\sqrt{1+v}} dv$$

$$D - 0 = 10 \int_0^{15} (1+v)^{1/2} - (1+v)^{-1/2} dv \quad \checkmark$$

$$D = \left[\frac{(1+v)^{3/2}}{3/2} - \frac{(1+v)^{1/2}}{1/2} \right]_0^{15}$$

$$= \frac{2}{3} (16\sqrt{16} - 1\sqrt{1}) - 2(\sqrt{16} - \sqrt{1})$$

$$\therefore D = 360 \text{ m} \quad \checkmark$$

$$(c)(i) \quad \frac{5}{(x+3)(2x+1)} = \frac{A}{x+3} + \frac{B}{2x+1}$$

$$5 = A(2x+1) + B(x+3)$$

$$\text{Sub } x = -\frac{1}{2}$$

$$5 = A(0) + B\left(\frac{5}{2}\right)$$

$$B = 2$$

$$\text{Sub } x = -3$$

$$5 = A(-5) + B(0)$$

$$A = -1$$

$$\therefore A = -1, B = 2$$

$$(ii) \quad \int_0^2 \frac{5}{(x+3)(2x+1)} dx = \int_0^2 \frac{-1}{x+3} + \frac{2}{2x+1} dx$$

$$= \left[-\ln|x+3| + \ln|2x+1| \right]_0^2$$

$$= \left[\ln \left| \frac{2x+1}{x+3} \right| \right]_0^2$$

$$= \ln\left(\frac{5}{3}\right) - \ln\left(\frac{1}{3}\right)$$

$$= 0 - \ln\left(\frac{1}{3}\right)$$

$$= -\ln\left(\frac{1}{3}\right)$$

$$= \ln 3$$

$$(d) \quad S: \left| \vec{r} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right| = 3 \quad \text{and} \quad L: \vec{r} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

POI when

$$\left| \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right| = 3$$

$$\left| \begin{pmatrix} 1+\lambda \\ 1+\lambda \\ -1+2\lambda \end{pmatrix} \right| = 3$$

$$|\underline{a}|^2 = \underline{a} \cdot \underline{a}$$



$$\begin{pmatrix} 1+\lambda \\ 1+\lambda \\ -1+2\lambda \end{pmatrix} \cdot \begin{pmatrix} 1+\lambda \\ 1+\lambda \\ -1+2\lambda \end{pmatrix} = 9$$

$$(1+\lambda)^2 + (1+\lambda)^2 + (-1+2\lambda)^2 = 9$$

$$1+2\lambda+\lambda^2 + 1+2\lambda+\lambda^2 + 1-4\lambda+4\lambda^2 = 9$$

$$6\lambda^2 = 6$$

$$\lambda = \pm 1$$



Sub $\lambda = 1$ in L :

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

Sub $\lambda = -1$ in L :

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - 1 \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$

✓ Both POIs

Question 14

(a) (i) $x = A \sin(\omega t + \alpha)$

$$A = 6$$

$$T = \frac{2\pi}{\omega} = 16$$

$$\omega = \frac{\pi}{8}$$

✓

$$\text{At } t=2, x=6$$

$$6 = 6 \sin\left(\frac{\pi}{4} + \alpha\right)$$

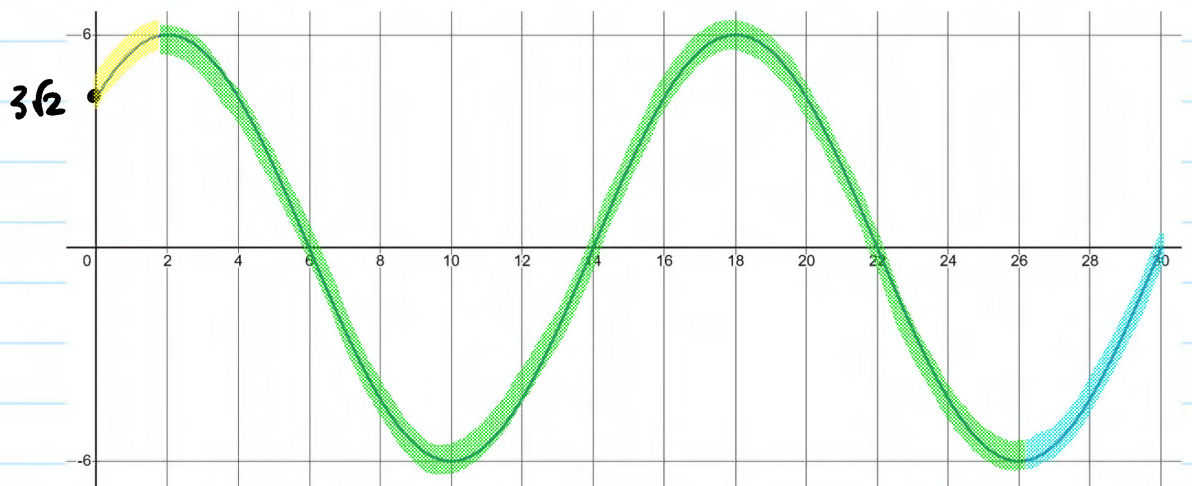
$$\frac{\pi}{4} + \alpha = \sin^{-1}(1) = \frac{\pi}{2}$$

$$\alpha = \frac{\pi}{4}$$

$$\therefore x = 6 \sin\left(\frac{\pi t}{8} + \frac{\pi}{4}\right)$$

✓

(ii)



$$\text{At } t=0, x = 6 \sin\left(0 + \frac{\pi}{4}\right) = \frac{6}{\sqrt{2}} = 3\sqrt{2}$$

✓

$$\text{Distance} = (6 - 3\sqrt{2}) + 3 \times 12 + 6$$

$$= 48 - 3\sqrt{2} \text{ metres}$$

✓

$$(b) (i) \text{ RHP: } \tan 4\theta = \frac{4t(1-t^2)}{1-6t^2+t^4}$$

$$\text{LHS} = \tan 4\theta$$

$$= \tan(2 \times 2\theta)$$

$$= \frac{2 \tan 2\theta}{1 - \tan^2 2\theta}$$

$$= \frac{2 \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right)}{1 - \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right)^2}$$

Sub $t = \tan \theta$



$$= \frac{\frac{4t}{1-t^2}}{1 - \frac{4t^2}{(1-t^2)^2}}$$

$$= \frac{\frac{4t}{1-t^2}}{\frac{1-2t^2+t^4-4t^2}{(1-t^2)^2}}$$

$$= \frac{4t(1-t^2)}{1-6t^2+t^4}$$



$$= \text{RHS}$$

$$(ii) \tan \theta \tan 4\theta = 1$$

$$t \times \frac{4t(1-t^2)}{1-6t^2+t^4} = 1$$



$$4t^2(1-t^2) = 1-6t^2+t^4$$

$$4t^2 - 4t^4 = 1-6t^2+t^4$$



$$\therefore 5t^4 - 10t^2 + 1 = 0$$

$$(iii) \quad t^2 = \frac{10 \pm \sqrt{80}}{2(5)} = \frac{5 \pm 2\sqrt{5}}{5}$$

$$t = \pm \sqrt{\frac{5 \pm 2\sqrt{5}}{5}}$$



$$\text{But } 0 < \tan \frac{\pi}{10} < \tan \frac{3\pi}{10}$$

$$\therefore \tan \frac{\pi}{10} = \sqrt{\frac{5-2\sqrt{5}}{5}}$$



(c)(i) IP: $\underbrace{\tan^{-1}\left(\frac{1}{k-1}\right)}_{\alpha} - \underbrace{\tan^{-1}\left(\frac{1}{k+1}\right)}_{\beta} = \tan^{-1}\left(\frac{2}{k^2}\right)$ for $k \geq 2$

Consider $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$
 $= \frac{\frac{1}{k-1} - \frac{1}{k+1}}{1 + \left(\frac{1}{k-1}\right)\left(\frac{1}{k+1}\right)}$

$\alpha = \tan^{-1}\left(\frac{1}{k-1}\right)$
 $\tan \alpha = \frac{1}{k-1}$
 $\beta = \tan^{-1}\left(\frac{1}{k+1}\right)$
 $\tan \beta = \frac{1}{k+1}$

$= \frac{\frac{k+1 - k+1}{(k-1)(k+1)}}{\frac{k^2 - 1 + 1}{(k-1)(k+1)}}$
 $= \frac{2}{k^2}$

$\alpha - \beta = \tan^{-1}\left(\frac{2}{k^2}\right)$

$\therefore \tan^{-1}\left(\frac{1}{k-1}\right) - \tan^{-1}\left(\frac{1}{k+1}\right) = \tan^{-1}\left(\frac{2}{k^2}\right)$

(ii) $\sum_{k=1}^{\infty} \tan^{-1}\left(\frac{2}{k^2}\right)$

$= \tan^{-1} 2 + \sum_{k=2}^{\infty} \tan^{-1}\left(\frac{1}{k-1}\right) - \tan^{-1}\left(\frac{1}{k+1}\right)$

$= \tan^{-1} 2 + \underbrace{\tan^{-1}(1) - \tan^{-1}\left(\frac{1}{2}\right)}_{k=2} + \underbrace{\tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{4}\right)}_{k=3} + \underbrace{\tan^{-1}\left(\frac{1}{4}\right) - \tan^{-1}\left(\frac{1}{6}\right)}_{k=4}$
 $+ \underbrace{\tan^{-1}\left(\frac{1}{6}\right) - \tan^{-1}\left(\frac{1}{8}\right)}_{k=5} + \dots + \underbrace{\tan^{-1}\left(\frac{1}{n-2}\right) - \tan^{-1}\left(\frac{1}{n}\right)}_{k=n-1} + \underbrace{\tan^{-1}\left(\frac{1}{n-1}\right) - \tan^{-1}\left(\frac{1}{n+1}\right)}_{k=n}$

$= \tan^{-1} 2 + \frac{\pi}{4} + \tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{n}\right) - \tan^{-1}\left(\frac{1}{n+1}\right)$

$\lim_{n \rightarrow \infty} \left[\sum_{k=1}^n \tan^{-1}\left(\frac{2}{k^2}\right) \right] = \lim_{n \rightarrow \infty} \left(\frac{\pi}{4} + \tan^{-1} 2 + \tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{n}\right) - \tan^{-1}\left(\frac{1}{n+1}\right) \right)$

$= \frac{\pi}{4} + \tan^{-1} 2 + \tan^{-1}\left(\frac{1}{2}\right) - \lim_{n \rightarrow \infty} \left(\tan^{-1}\left(\frac{1}{n}\right) + \tan^{-1}\left(\frac{1}{n+1}\right) \right)$

$= \frac{\pi}{4} + \frac{\pi}{2} - 0$ $\tan(A+B) = \frac{2 + \frac{1}{2}}{1 - (2 \times \frac{1}{2})} = \text{undefined}$

$= \frac{3\pi}{4}$

OR... $\tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2}, x > 0$

Question 15

$$(a) \vec{OM} = \frac{1}{2} \vec{OA} = \frac{1}{2} \underline{a}$$

$$\vec{OE} = \lambda \vec{OA} = \lambda \underline{a} \quad \lambda \text{ scalar}$$

$$\vec{ON} = \frac{1}{2} \vec{OB} = \frac{1}{2} \underline{b}$$

$$\vec{OF} = \mu \vec{OB} = \mu \underline{b} \quad \mu \text{ scalar}$$

$$\text{Now, } |\vec{OM}| |\vec{OE}| = \frac{\lambda}{2} |\underline{a}|^2$$

$$|\vec{ON}| |\vec{OF}| = \frac{\mu}{2} |\underline{b}|^2$$

$$\text{Also, } AF \perp OB, \text{ so } \vec{AF} \cdot \vec{OB} = 0$$

$$(\mu \underline{b} - \underline{a}) \cdot \underline{b} = 0$$

$$\mu \underline{b} \cdot \underline{b} - \underline{a} \cdot \underline{b} = 0$$

$$\underline{a} \cdot \underline{b} = \mu \underline{b} \cdot \underline{b} = \mu |\underline{b}|^2$$

$$\text{Similarly, } BE \perp OA, \text{ so } \vec{BE} \cdot \vec{OA} = 0$$

$$(\underline{b} - \lambda \underline{a}) \cdot \underline{a} = 0$$

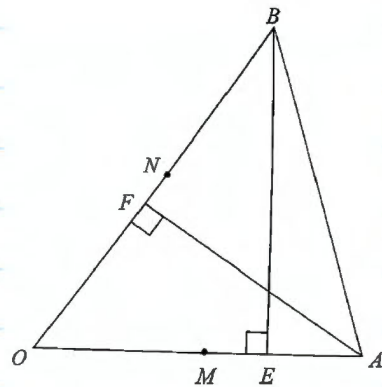
$$\underline{a} \cdot \underline{b} - \lambda \underline{a} \cdot \underline{a} = 0$$

$$\underline{a} \cdot \underline{b} = \lambda \underline{a} \cdot \underline{a} = \lambda |\underline{a}|^2$$

$$\text{Then } \underline{a} \cdot \underline{b} = \lambda |\underline{a}|^2 = \mu |\underline{b}|^2$$

$$\frac{\lambda}{2} |\underline{a}|^2 = \frac{\mu}{2} |\underline{b}|^2$$

$$\therefore |\vec{OM}| |\vec{OE}| = |\vec{ON}| |\vec{OF}|$$



$$(b) \quad I_n = \int_0^1 \frac{x^n}{x^2+1} dx$$

$$(i) \quad LHS = I_n + I_{n-2}$$

$$= \int_0^1 \frac{x^n}{x^2+1} dx + \int_0^1 \frac{x^{n-2}}{x^2+1} dx$$

$$= \int_0^1 \frac{x^{n-2} (x^2+1)}{x^2+1} dx$$

$$= \left[\frac{x^{n-1}}{n-1} \right]_0^1$$

$$= \frac{1}{n-1} - 0$$

$$= \frac{1}{n-1}$$

$$= RHS$$

$$(ii) \quad \int_0^1 \frac{x^2(x-1)^2}{x^2+1} dx = \int_0^1 \frac{x^2(x^2-2x+1)}{x^2+1} dx$$

$$= \int_0^1 \frac{x^4}{x^2+1} - 2 \frac{x^3}{x^2+1} + \frac{x^2}{x^2+1} dx$$

$$= I_4 - 2I_3 + I_2$$

$$= \frac{1}{3} - 2\left(\frac{1}{2} - I_1\right) \quad \text{from (i)}$$

$$= \frac{1}{3} - 1 + \int_0^1 \frac{2x}{x^2+1} dx$$

$$= -\frac{2}{3} + \left[\ln(x^2+1) \right]_0^1$$

$$= -\frac{2}{3} + \ln 2 - \ln 1$$

$$= \ln 2 - \frac{2}{3}$$

$$(i) f_n(x) = e^x - \left(1 + \sum_{r=1}^n \frac{x^r}{r!}\right) \quad n=1, 2, 3, \dots$$

$$(ii) f_n(0) = e^0 - \left(1 + \sum_{r=1}^n \frac{0^r}{r!}\right)$$

$$= 1 - 1 - 0$$

$$= 0$$

$$f_{n+1}(x) = e^x - \left(1 + \sum_{r=1}^{n+1} \frac{x^r}{r!}\right)$$

$$= e^x - 1 - \left(\frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots + \frac{x^n}{n!} + \frac{x^{n+1}}{(n+1)!}\right)$$

$$f_{n+1}'(x) = e^x - 0 - \left(\frac{1}{1!} + \frac{2x}{2!} + \frac{3x^2}{3!} + \frac{4x^3}{4!} + \dots + \frac{nx^{n-1}}{n!} + \frac{(n+1)x^n}{(n+1)!}\right)$$

$$= e^x - \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^{n-1}}{(n-1)!} + \frac{x^n}{n!}\right)$$

$$= e^x - \left(1 + \sum_{r=1}^n \frac{x^r}{r!}\right)$$

$$= f_n(x)$$

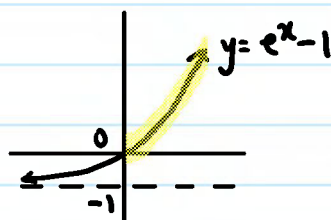
$$(ii) f_1(x) = e^x - \left(1 + \frac{x^1}{1!}\right)$$

$$= e^x - 1 - x$$

$$f_1'(x) = e^x - 0 - 1$$

$$= e^x - 1$$

$$> 0 \text{ for } x > 0$$



$\therefore f_1(x)$ is increasing for $x > 0$

Since $f_1(0) = 0$, then $f_1(x) > 0$ for all $x > 0$

$$e^x - 1 - x > 0$$

$$\therefore 1 + x < e^x \text{ for all } x > 0$$

(iii) RTP: $1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} < e^x$ for $x > 0$.

Step 1 For $n=1$

$$\text{LHS} = 1+x$$

$$\text{RHS} = e$$

$$[1+x < e^x \text{ from (ii)}]$$

LHS < RHS, statement is true for $n=1$ ✓

Step 2 Assume statement is true for $n=k$

$$\text{i.e. } 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^k}{k!} < e^x$$

$$\Rightarrow e^x - \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^k}{k!}\right) = f_k(x) > 0 \quad \checkmark$$

Step 3 Attempt to prove statement is true for $n=k+1$

$$\text{i.e. } 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^{k+1}}{(k+1)!} < e^x$$

$$\text{RHS} - \text{LHS} = e^x - \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^{k+1}}{(k+1)!}\right)$$

$$= f_{k+1}(x)$$

$$> 0, \text{ since } f_{k+1}(0) = 0$$

$$\text{and } f_{k+1}'(x) = f_k(x) > 0 \quad \text{(i) and from assumption} \quad \checkmark$$

$$\therefore \text{RHS} > \text{LHS}$$

$\therefore$ Statement is true for $n=k+1$ if it is true for $n=k$

Step 4 By the principle of mathematical induction, the statement is true for integers $n \geq 1$.

Question 16

(a) (i) RP: $(1+i\tan\theta)^n + (1-i\tan\theta)^n = \frac{2\cos n\theta}{\cos^n\theta}$ $\cos\theta \neq 0, n \in \mathbb{Z}^+$

$$\text{LHS} = (1+i\tan\theta)^n + (1-i\tan\theta)^n$$

$$= \left(1 + \frac{i\sin\theta}{\cos\theta}\right)^n + \left(1 - \frac{i\sin\theta}{\cos\theta}\right)^n$$

$$= \left(\frac{\cos\theta + i\sin\theta}{\cos\theta}\right)^n + \left(\frac{\cos\theta - i\sin\theta}{\cos\theta}\right)^n$$

$$= \left(\frac{\text{cis}\theta}{\cos\theta}\right)^n + \left(\frac{\text{cis}(-\theta)}{\cos\theta}\right)^n$$

$$= \frac{\text{cis}(n\theta) + \text{cis}(-n\theta)}{\cos^n\theta}$$

$$= \frac{\cos(n\theta) + i\sin(n\theta) + \cos(n\theta) - i\sin(n\theta)}{\cos^n\theta}$$

$$\begin{aligned}\cos(-n\theta) &= \cos(n\theta) \\ \sin(-n\theta) &= -\sin(n\theta)\end{aligned}$$

$$= \frac{2\cos(n\theta)}{\cos^n\theta}$$

$$= \text{RHS}$$

(ii) Let $z = i\tan\theta, n=2$

$$(1+z)^2 + (1-z)^2 = 0 \rightarrow \text{Quadratic eqn, 2 roots}$$

$$\frac{2\cos 2\theta}{\cos^2\theta} = 0$$

$$\cos 2\theta = 0$$

$$2\theta = \pi/2, 3\pi/2$$

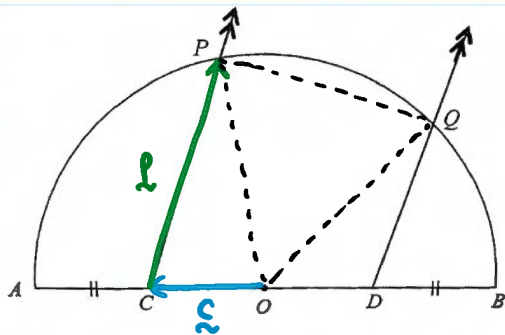
$$\theta = \pi/4, 3\pi/4$$

$$z_1 = i\tan\frac{\pi}{4}$$

$$z_2 = i\tan\frac{3\pi}{4} = i\tan\left(-\frac{\pi}{4}\right) = -i\tan\frac{\pi}{4}$$

$$\therefore z = \pm i\tan\frac{\pi}{4}$$

(b)



$$(i) \quad \vec{DQ} = \lambda \vec{CP} \quad (\text{DQ} \parallel \text{CP})$$

$$\therefore \vec{DQ} = \lambda \vec{p} \quad \text{for } \lambda > 0 \text{ scalar}$$

✓ Explain

$$\begin{aligned} \text{Now, } \vec{OP} &= \vec{OC} + \vec{CP} \\ &= \vec{s} + \vec{p} \\ \vec{OQ} &= \vec{OD} + \vec{DQ} \\ &= -\vec{s} + \lambda \vec{p} \end{aligned}$$

Then, $|\vec{OP}| = |\vec{OQ}|$ (equal radii of circle)

$$(\vec{s} + \vec{p}) \cdot (\vec{s} + \vec{p}) = (-\vec{s} + \lambda \vec{p}) \cdot (-\vec{s} + \lambda \vec{p})$$

$$\cancel{s} \cdot \cancel{s} + 2\vec{s} \cdot \vec{p} + \vec{p} \cdot \vec{p} = \cancel{s} \cdot \cancel{s} - 2\lambda \vec{s} \cdot \vec{p} + \lambda^2 \vec{p} \cdot \vec{p}$$

$$\vec{p} \cdot \vec{p} (1 - \lambda^2) = -2\vec{s} \cdot \vec{p} (1 + \lambda)$$

$$\vec{p} \cdot \vec{p} (1 - \lambda)(1 + \lambda) = -2\vec{s} \cdot \vec{p} (1 + \lambda)$$

$$\therefore (1 - \lambda) \vec{p} \cdot \vec{p} + 2\vec{s} \cdot \vec{p} = 0$$

$$(ii) \quad \vec{PQ} = \vec{PC} + \vec{CB} + \vec{DQ}$$

$$= -\vec{p} - 2\vec{s} + \lambda \vec{p}$$

$$= (\lambda - 1) \vec{p} - 2\vec{s}$$

$$\text{Now, } \vec{CP} \cdot \vec{PQ} = \vec{p} \cdot ((\lambda - 1) \vec{p} - 2\vec{s})$$

$$= (\lambda - 1) \vec{p} \cdot \vec{p} - 2\vec{s} \cdot \vec{p}$$

$$= -[(1 - \lambda) \vec{p} \cdot \vec{p} + 2\vec{s} \cdot \vec{p}]$$

$$= 0$$

from (i)

$$\therefore \text{CP} \perp \text{PQ}$$

$$\therefore \angle CPQ = 90^\circ$$

$$(c)(i) \quad I_n = \int_1^e (1 - \ln x)^n dx$$

$$\begin{aligned} u &= (1 - \ln x)^n & v' &= 1 \\ u' &= \frac{n(1 - \ln x)^{n-1}}{-x} & v &= x \end{aligned}$$

$$= \left[x(1 - \ln x)^n \right]_1^e - \int_1^e \frac{n(1 - \ln x)^{n-1}}{-x} \cdot x dx$$

$$= e(1 - \ln e)^n - 1(1 - \ln 1)^n + n \int_1^e (1 - \ln x)^{n-1} dx$$

$$= 0 - 1 + n I_{n-1}$$

$$= -1 + n I_{n-1}$$

$$(ii) \text{ PTP: } I_n = n!e - 1 - \sum_{r=1}^n n P_r \quad n \geq 1$$

Step 1 for $n=1$

$$\text{LHS} = I_1$$

$$= -1 + I_0$$

$$= -1 + \int_1^e dx$$

$$= -1 + e - 1$$

$$= e - 2$$

$$\text{RHS} = 1!e - 1 - {}^1P_1$$

$$= e - 1 - 1$$

$$= e - 2$$

$\therefore \text{LHS} = \text{RHS}$, Statement is true for $n=1$

Step 2 Assume statement is true for $n=k$

$$\text{i.e. } I_k = k!e - 1 - \sum_{r=1}^k k P_r$$

Step 3 Attempt to prove true for $n=k+1$

$$\text{i.e. } I_{k+1} = (k+1)!e - 1 - \sum_{r=1}^{k+1} (k+1) P_r$$

$$\text{LHS} = I_{k+1}$$

$$= -1 + (k+1) I_k$$

$$= -1 + (k+1) \left[k!e - 1 - \sum_{r=1}^k k P_r \right]$$

$$= -1 + (k+1)!e - (k+1) - (k+1) [{}^kP_1 + {}^kP_2 + {}^kP_3 + \dots + {}^kP_k]$$

PTD $\rightarrow$

$$= -1 + (k+1)!e - (k+1) - (k+1) \left[\frac{k!}{(k-1)!} + \frac{k!}{(k-2)!} + \frac{k!}{(k-3)!} + \dots + \frac{k!}{(k-k)!} \right]$$

$$= -1 + (k+1)!e - (k+1) - \left(\frac{(k+1)!}{(k+1-2)!} + \frac{(k+1)!}{(k+1-3)!} + \frac{(k+1)!}{(k+1-4)!} + \dots + \frac{(k+1)!}{(k+1-(k+1))!} \right)$$

$$= -1 + (k+1)!e - (k+1) - \sum_{r=2}^{k+1} (k+1)p_r$$

$$= -1 + (k+1)!e - (k+1) - \left(\sum_{r=1}^{k+1} (k+1)p_r - (k+1)p_1 \right)$$

$$= -1 + (k+1)!e - \cancel{(k+1)} - \sum_{r=1}^{k+1} (k+1)p_r + \cancel{(k+1)}$$

$$= (k+1)!e - 1 - \sum_{r=1}^{k+1} (k+1)p_r$$

= RHS

∴ Statement is true for $n=k+1$ if it is true for $n=k$

Step 4 By the principle of mathematical induction, statement is true for integers $n \geq 1$.